Walter Sisulu University
In pursuit of excellence

STUDENT MODULE GUIDE

SMG CSS37W1/1/2026

CSS37W1 Software Engineering I

Faculty of Natural Sciences

Excellence In Teaching & Learning

IMPORTANT INFORMATION

NB: Each student registered for this module is entitled to 1 (one) copy of this Student Module Guide (SMG). Students are advised, therefore, to keep their copy in a safe place, in case it is required in future when registering and applying for credits at other institutions.

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1. Introduction & Welcome

Software Engineering: Crafting the Future, One Line of Code at a Time

Software engineering isn't just about writing code; it's about building the future. It's a discipline that draws upon a rich tapestry of knowledge, weaving together computer science fundamentals with principles from mathematics, engineering, project management, and even diverse application domains. Imagine the possibilities: from revolutionizing healthcare with cutting-edge medical software to designing the next generation of smart cities, software engineers are the architects of the digital world.

This journey into software engineering will equip you with the skills to transform abstract ideas into tangible realities. You'll learn to bridge the gap between theory and practice, mastering the art of abstraction and modeling – essential tools for tackling complex challenges. We'll explore how to acquire specialized domain knowledge, enabling you to build software solutions that truly meet the needs of specific industries and users. Above all, you'll discover the profound impact of elegant and robust design, the cornerstone of any successful software project.

Our course is structured to guide you through the entire software development lifecycle, from initial concept to final deployment. We'll begin with a broad overview of software engineering principles, laying the groundwork for a deeper dive into each critical phase of a software project. You'll gain a comprehensive understanding of the current state of software development, exploring established best practices and emerging trends.

Software engineering sits at the intersection of computer science and problem-solving. Just as computer scientists conduct research to advance our understanding of computing, software engineers apply these findings to create practical tools and techniques. We're not just coders; we're problem solvers, using our knowledge of computers and computation to address real-world challenges. This process involves both analysis – dissecting complex problems into manageable components – and synthesis – constructing innovative solutions from these carefully analysed pieces. Throughout this course, you'll become familiar with a diverse array of methods, tools, procedures, and paradigms that empower

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software engineers to tackle any project with confidence. Get ready to embark on a journey of creation, innovation, and problem-solving – the world of software engineering awaits!

The module will take a modern approach to software engineering with focus on software product development which looks at software development from product-based instead of project-based engineering. This module should take about 13 weeks to complete. You can check out the recommended module schedule to see a quick overview of the lessons and assignments you'll complete each week.

The purpose of this Student Module Guide is to provide you with written guidelines as to what to expect from this module and what we, in turn, expect of you regarding this module.

2. Module Purpose & Outcomes

2.1 Module Purpose

The purpose of this module is to provide students with the knowledge and technical skills required to analyse, design, implement and maintain software systems. It is also intended to provide students with competence in the understanding and appreciation of the field of software engineering.

This course is for undergraduate students majoring in computer science who have passed the prerequisites and for any other students who meets the requirements from other programmes. Students will be able to follow the software product development process which will result into a demonstrated software product in semester II. Software engineering relies on many areas in computer science for its theoretical and conceptual foundations, but it also requires students to utilize concepts from a variety of other fields, such as mathematics, engineering, and project management, and one or more application domains. Students will be

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expected to able to integrate theory and practice, to recognize the importance of abstraction and modelling, to be able to acquire special domain knowledge beyond the computing discipline for the purposes of supporting software development in specific domains of application, and to appreciate the value of good design.

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2.2 Module Learning Outcomes & Assessment Criteria

Module Learning Outcomes **THE SUCCESSFUL STUDENT WILL BE EXPECTED TO…)**	**Module Assessment Criteria**
Present the general overview of the field of software engineering and professional practice.	• Clearly articulates the concept of software engineering, highlighting distinctions from computer science. • Identifies and elucidates key principles and concepts integral to software engineering. • Examines the historical track record of software engineering, critically assessing its successes and challenges. • Discusses how an engineering approach effectively manages risk and uncertainty in the context of software development. • Evaluates the criteria for defining good software, demonstrating an understanding of quality metrics. • Summarizes the roles and responsibilities of stakeholders within a software team, demonstrating an ability to synthesize and present key information concisely.

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Apply core software engineering practices at conceptual level for a given problem.	• Outline different software processes. • Interpret the given situation and recommend the missing engineering practices causing that situation. • Select the appropriate engineering practices for a given situation. • Describe the key engineering practices and their purpose

critically examine and assess various software development methodologies, considering their appropriateness in different contexts.	• Compare the traditional waterfall model to the agile model, highlighting their distinctive features and assessing the advantages and disadvantages of each. • Discuss agile methods, identify associated issues, and formulate solutions to mitigate these issues effectively.

Propose a methodology best suited for a given situation.	• Explain the software lifecycle and its various phases, emphasizing the interdependencies between phases. • Describe the deliverables produced at each phase and apply this understanding to a practical context. • Apply an appropriate agile model to solve a given problem, demonstrating the practical application of agile principles in a specific context.

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Apply the concepts of software products and software product management in modern software development process.	• Articulate the definition of a software product and elucidate the responsibilities of a software product manager. • Distinguish between project-based software engineering and product-based software engineering. • Evaluate software platforms, software execution models and software product vision. • Demonstrate the application of project management skills by developing a project plan for a software product. This plan should encompass change management, resource allocation, as well as the identification and management of project risks. • Develop Software product prototyping
Demonstrate detailed understanding of the modern approach capturing and recording software requirements in a project.	• Create personas that vividly represent diverse user types, considering demographics, behaviours, and goals. • Develop a product vision from a real-world problem. • Develop comprehensive scenarios covering a range of user tasks and potential situations.
	• Integrate personas, scenarios, user stories, and feature identification cohesively to guide product development. • Apply software features, scenarios, and stories to selected project.

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Explore the role of software architecture in the software product development.	• Describe what constitutes software architecture. • Analyse how software architecture affects non-functional requirements. • Describe and evaluate how software systems are decomposed. • Outline different ways of how architectures are distributed. • Explore technologies issues when deciding software architecture. Recommend software architectures given software requirements.
Demonstrate detailed knowledge of how to apply the modern approach of how software products are delivered via the cloud	• Explain what cloud and cloud computing is. • Describe virtualization and containers Illustrate various types of everything as a • service Elaborate how multitenant and Multiple instances are implemented in the cloud.

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Highlight the potential of Generative AI in software engineering	• Apply generative AI tools (e.g., GitHub Copilot, ChatGPT) to enhance software engineering tasks such as code generation, debugging, testing, and documentation.

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2.3 Module NQF Level

NQF Level 7

2.4 Pre – and Co-requisites

For mainstream students ensure that you passed the following

- CSI26W1 PROGRAMMING IN JAVA
- CSI26W2 OPERATING SYSTEMS
- CSP26W1 DATA COMMUNICATION & NETWORKING
- CSP26W2 DATA STRUCTURES

2.5 Module Credits

14 Credits

2.6 Notional Hours Required

140 Notional hours. This module is on level 7 of the NQF (National Qualifications

Framework) and it contributes 14 credits towards your final qualification. According to the Department of Higher Education, you should therefore spend 140 notional hours completing this module; these hours include time spent **studying**, doing the **online activities**, completing **assignments** and writing the exam. The module is run over a semester of 14 weeks, and 3 hours is deducted from the notional hours for the exam. The time you will need to spend on this module will thus be as follows: 137 (hours) ÷ 14 (weeks) = 9.79 hours per week per module 9.79 (hours) ÷ 7 (days) = 1,4 hours per day per module

3. Qualification & Programme Information

3.1 Qualification Rules

The Software Engineering I module is part of the BSc Computer Science programme. It's a service module for other qualifications such BSc Applied Mathematics, BSc Applied Statical Science, BSc Mathematics and BSc Physics. For details of the qualification rules check the FNS Prospectus.

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In general, in order to qualify you must pass all the core modules of each qualification along with the required elective modules. You will be expected to obtain a minimum of 360 credits in order to graduate.

3.2 Qualification Title

BSc in Computer Science

3.3 Qualification NQF level

NQF Level 7

3.4 Qualification Total credits

360 Credits

3.5 Mode of Delivery & Access to Learner Management System

Walter Sisulu university has adopted blended learning approach which is a mix of traditional face to face learning and online learning.

Moodle, a Learning Management System (LMS) designed for various teaching and learning goals, has been introduced as one of the E-learning platforms used by WSU.

Some other platforms such as Zoom, and Microsoft Teams may also be used to supplement lack of face-to-face interaction.

In order to access the Moodle platform, follow the link below:

1. URL Address – https://wiseup.wsu.ac.za/
2. Enter username (your student number) and password (usually your ID number) and when prompted.
3. Click Login Button

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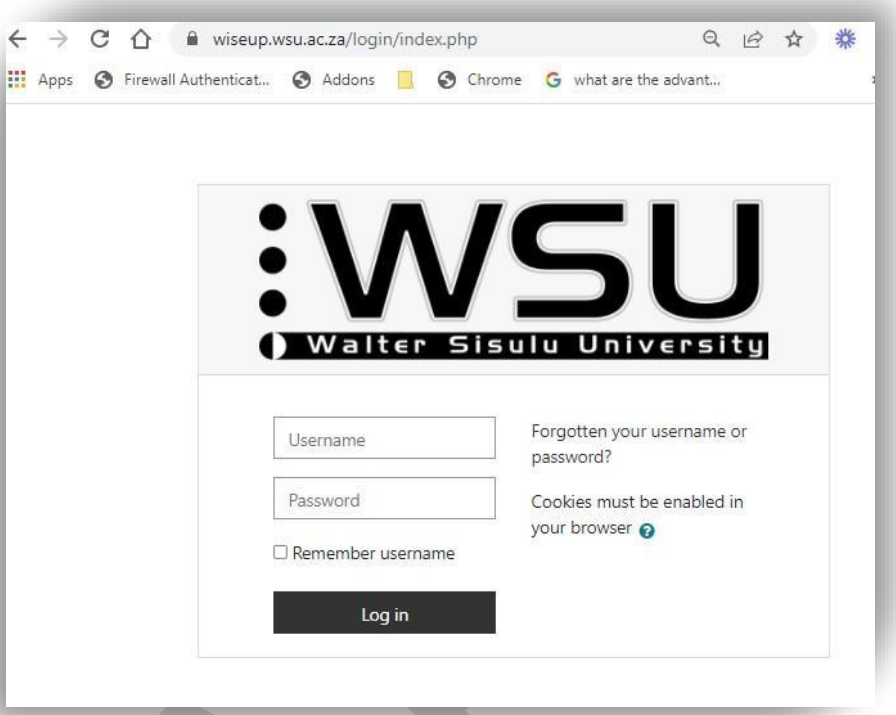

Microsoft Teams is used for conducting of Online Live lectures and demos. You can join the class by visiting the link:

Link: General | CSS37W1 - Software Engineerng I 2026 | Microsoft Teams

WhatsApp Group to be used for quick sending of announcements. Please note that this platform will not be used for teaching. Click on the link WhatsApp group.

- https://chat.whatsapp.com/DUpNFAQekgzCnhfg7eAqCw?mode=gi_t

4. Contact with the Lecturers, Department and Faculty

The following are contact details of various Units within the university where you can get help.

4.1 Communication with Lecturers

If you have any academic (module-related) questions, I will be glad to answer them as soon as possible. Please feel free to ask as many questions as you wish.

You could place your questions on the Discussion forums on WiSeUp, Teams, or the WhatsApp Group. In this way, your fellow students can benefit from the answers you receive.

If you do not feel comfortable posting your question in the discussion forums, WhatsApp Group, or Teams, do not hesitate to send an email.

You are also welcome to make an appointment with me to discuss any academic issues you may encounter while working through your study material. When making an appointment, please specify which section of the work you need to be clarified. I may not assist students with general enquiries.

Office hours are 08:00 to 16:30, Monday to Thursday, and 08:00 to 15:30 on Friday.

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Timetable:	**Lecturer**:
• Monday: 08:45 – 10:15 (ES2) • Tuesday: 14:00 - 15:30 (EL5) • Friday: 13:15 - 14:45 (EL16)	Mr. V. O. Adeniji Lecturer – Mathematical Sciences and Computing Faculty of Natural Sciences Tel: | Cell: E-mail: vadeniji@wsu.ac.za

4.2 Contacting the Department

Although your lecturer would like to help with all your questions, some things are just beyond my abilities. Kindly contact the department directly, should you have any questions related to the topics listed below: -

- Cancellation of Modules, Addition of Modules, Marks not appearing, PGs not showing.

Department Details

Acting Head of Department	**Departmental Secretary**
Dr. R. Panicker Senior Lecturer – Mathematical Sciences and Computing Faculty of Natural Sciences Tel: 047-5022230 | Cell: Office TS025 East Teaching Mall, Second Floor, NMD Campus. E-mail: rpanicker@wsu.ac.za	Mrs Tandekile Nkohla Faculty of Natural Sciences Tel: | Cell: Office TS027 East Teaching Mall, Second Floor, NMD Campus. E-mail: tnkohla@wsu.ac.za

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4.3 Contacting the University

When contacting the university, you are required to use your myWSU email address; emails from any other address may not be answered because they land in junk mail, which is sometimes automatically deleted. It is easier to identify you simply by your WSU email address. It further gives the university confidence that we are communicating with a registered student.

For details on who to contact for these issues, please visit the following website for email addresses and phone numbers:

Faculty of Natural Sciences

Executive Dean: Professor A. Modi | amodi@wsu.ac.za | 060 531 0451 |Ext 2454

Faculty Administrator: Ms N Dangazele | ndangazele@wsu.ac.za | Ex 1968

Teaching Development: Mr A. Mapfumo | amapfumo@wsu.ac.za | Ext 2304

University Library Services

- https://wsu-ac.libguides.com/services
- Faculty of Natural Sciences Librarian
  - Ms Bathandiwe Bingwa
  - Phone: 0475022307 | Email: bbingwa@wsu.ac.za

Examination Matters

- Ms A. Jiba – The Chief Examination Officer | ajiba@wsu.ac.za

Learning Management Systems

- Mr L.S. Cwala (lscwala@wsu.ac.za) - 067 805 5267

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5. Resources

The following sections explain the resources and the platform where these resources will be made available.

5.1 Study Material

Study materials will be posted on the WiSeUp eLearning Platform (Moodle).

5.2 Recommended Material

1. Sommerville, I. (2020). Engineering software products: an introduction to modern software engineering. Pearson.
2. Shari Lawrence Pfleeger and Atlee, J.M. (2010). Software engineering: theory and practice. Upper Saddle River N.J.: Prentice Hall.
3. Braude, Bernstein: Software Engineering: Modern Approaches, 2nd Edition
4. Pressman Rodgers Software Engineering A Practitioner Approach 9th Edition
5. McLaughlin B.D, Pollice G,West D. Head First Object-Oriented Analysis and Design A Brain Friendly Guide to OOA&D (2006)
6. Dan Pilone and Russ Miles (2008), HeadFirst Software Development, 1st Edition, 2008 O'Reilly Media

5.3 Electronic Resources / Material

1. YouTubes. https://youtube.com/playlist?list=PL_pbwdIyffslgxMVyXhnHiSn_EWTvx1G-&si=jvPeDtpcHdxU2yZA – Software Engineering Full Course.

5.4 Other Resources

1. https://www.coursera.org/learn/software-processes

2. https://www.coursera.org/learn/introduction-to-software-engineering (Introduction to Software Engineering | Coursera)

5.5 Accessing the Library

Details about accessing the library.

- https://www.wsu.ac.za/index.php/university-library

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6 Student Support Services

For student support services, use the following information.

6.1 Student Counselling

Counselling Unit: 047 502 2727

6.2 Tutor & PAL Services

Discussion classes with students will be available on Teams, WhatsApp, and the WiSeUp Discussion Forum.
There are no Tutor services in this module.

6.3 Writing Centre

Reading and writing centre: 047 502 2208 | rncube@wsu.ac.za

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7 Notes & Assignments

7.1 Class Notes

Class notes will be made available online via WiSeUp. Students are encouraged to visit WiSeUp and download the materials.

7.2 Assignments

CSS37W1 will be assessed through relevant assessment practices that will determine whether the required outcomes have been reached.

The formative assessments are designed and compiled to give a clear indication of whether you have grasped the outcomes of the module. The summative assessment, which will take place during the examination period, is done to ensure that you have grasped all the outcomes required to complete the module successfully. To complete this module successfully, you need to complete all assessments. The assessments are there for you to demonstrate what you have learnt up to a certain point. All the assessment results contribute towards your year mark.

All typed assessments must be submitted via the WiSeUp platform.

Quiz 1-5, Assignment, Practical Project, Test 1, and Test 2

8 Lecture Room Expectations

Since this module will be offered via the blended approach, online course discussions contribute to the development of your critical thinking and writing skills.

9 Rules for Online Engagement (Netiquette)

When contacting the university, you are required to use your WSU email account, e-mails received from any other e-mail address may not be answered. Since this

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module will be offered via the blended approach, course online discussions contribute to your critical thinking and writing skills' development. However, there is a netiquette code for online engagement too. In participating in online meeting, please follow the following rules.

- Use proper language.
- Be precise.
- Avoid emoticons and “texting” writing.
- Be explanatory. Justify your opinion.
- Read all comments before hitting “submit”.
- Tone down your language.
- Recognize and respect diversity.
- Control your temper.
- Take your posts seriously.
- Be credible.

Don’t type in UPPERCASE only, it is considered shouting and makes it difficult to read your posts.

http://elearningindustry.com/10-netiquette-tips-online-discussions

10 Formative Assessment

The following table provides a breakdown of the assessments, including weights and tentative due dates.

10.1 Assessment Plan

Assessment Number	**Type of Assessment**	**Weighting**	**Due Date**
1	Quiz 1	3%	TBA

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2	Quiz 2	3%	TBA
3	Quiz 3	3%	TBA
4	Test 1	25%	TBA
5	Quiz 4	3%	TBA
6	Quiz 5	3%	TBA
7	Assignment	10%	TBA
8	Test 2	25%	TBA
9	Practical Project	25%	TBA

10.2 Examination Requirements

10.2.1 DP Requirements

You are expected to minimum due performance (DP) mark of 40% to write the final examination.

10.2.2 Mark Calculation for DP

Quizzes 1-5 * 0.15 + Assignments * 0.10 + Test 1 * 0.25 + Test 2 * 0.25 + Project * 0.25

11 Summative Assessment

11.1 Examination Period

May/June 2026

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11.2 Examination Paper & Admission to the Examination

The examination paper will be out of 100 marks. There will be five questions covering the whole syllabus. The examination will be venue-based. The duration of the paper will be at least 3 hours. To qualify to write the examination, you must have a DP mark of at least 40%.

11.3 Marking of Examination Papers

Examination will be marked based on the prepared marking key and rubric.

12 Assessments: Compulsory Assessments

12.1 Assessment 01 – Semester 1

Assignment 1.

12.1.1 Instructions & Due Date

This assignment expects you to complete two short courses and obtain a badge. You will be expected to upload the information to WiSeUp. Due Date: 12th April 2026. Not graded.

12.1.2 Assessment Questions

- Register at IBM Skills
- Enrol in the IBM Design Thinking Practitioner course.
  - Enterprise Design Thinking - Practitioner Course
- Register at IBM website. Enrol in the Git Course.
  - Badges: Git and GitHub Essentials - IBM Training - Global
  - IBM: Git and GitHub Basics | edX

12.2 Assessment 02 – Semester 1

Quiz 1-5

12.2.1 Instructions & Due Date

You will be expected to complete short quizzes weekly, which will be automatically graded. These will be available on Moodle.

Due dates: TBA

12.2.2 Assessment questions

Short questions covering the concepts covered in each Unit.

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12.3 Assessment 03 – Semester 1

Test 1

12.3.1 Instructions & Due Date

This Test will cover the first three topics covered in class.
Due dates: See the schedule above.

12.3.2 Assessment questions

Open-ended theory questions will be asked.

12.4 Assessment 04 – Semester 1

Test 2

12.4.1 Instructions & Due Date

This Test will cover the most concepts covered after the last three topics covered in class.

Due dates: See the schedule above.

12.4.2 Assessment questions

Open-ended questions covering issues of software architecture, requirements, and cloud-based software.

12.5 Assessment 05– Semester 1

Practical Project

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12.5.1 Instructions & Due Date

This semester's project focuses on understanding, documenting, and designing a software product. You will work in teams to analyse a given problem, elicit and document requirements, and create a preliminary design for a software application that addresses the problem. This document will form the foundation for the development phase in Semester 2. Creativity and application of software engineering principles learned in the course are essential.

Deliverable: A comprehensive written report detailing the project's requirements and initial design. This report should include architectural designs.

Team Formation: Teams should consist of 3-4 students.

Due Date: See the course schedule.

12.5.2 Assessment questions

This project is designed to give you practical experience in the early stages of the software development lifecycle, with a focus on requirements engineering and high-level design. Your team will engage in the followings:

Problem Analysis: Thoroughly investigate and understand the provided problem statement.

Requirements Elicitation: Employ appropriate techniques to elicit user and system requirements.

Requirements Documentation: Clearly and concisely document the gathered requirements, using appropriate formats and levels of detail. Consider functional, non-functional, and domain requirements.

Architectural Design: Develop a high-level software architecture for the proposed application, outlining key components, their interactions, and the chosen architectural style.

Design Rationale: Justify design decisions by linking them to the documented requirements.

This semester's work is crucial, as the documented requirements and designs will be used to develop a working application in Semester 2. Therefore, clarity, completeness, and accuracy are paramount.

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13 Module Content

13.1 Learning Unit 1: Introduction to Software Engineering

13.1.1 Key Concepts

What Is Software Engineering? The nature of software, Why Software Engineering, How Successful Have We Been? What Is Good Software? Who Does Software Engineering? A Systems

Approach, An Engineering Approach, Members of the Development Team, How Has

Software Engineering Changed? Legal and Ethical Issues in Computer

Science, Information Technology, and Software Engineering

13.1.2 Learning Outcomes and Assessment Criteria

Present the general overview of the field of software engineering and professional practice.

- Clearly articulates the concept of software engineering, highlighting distinctions from computer science.
- Identifies and elucidates key principles and concepts integral to software engineering.
- Examines the historical track record of software engineering, critically assessing its successes and challenges.
- Discusses how an engineering approach effectively manages risk and uncertainty in the context of software development.
- Evaluates the criteria for defining good software, demonstrating an understanding of quality metrics.
- Summarizes the roles and responsibilities of stakeholders within a software team, demonstrating an ability to synthesize and present key information concisely.
- Reference:
    - *Chapter 1 - Software Engineering Theory and Practice 4th Edition, Pearson Higher Education 2010*
    - *Chapter 1 - Software Engineering Practical Approach, 9th Edition*

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13.2 Learning Unit 2: Software Products

13.2.1 Key Concepts

Software, Software Products, Product Vision, Vision template, Product-based engineering, project-based engineering, software platforms, product management, software prototyping.

13.2.2 Learning Outcomes and Assessment Criteria

Apply the concepts of software products and software product management in modern software development process.

- Outline different software processes.
- Interpret the given situation and recommend the missing engineering practices causing that situation.
- Select the appropriate engineering practices for a given situation.
- Describe the key engineering practices and their purpose.
- Develop a Software product prototype.

Reference: Chapter 2 Software Products – Ian Sommerville – Engineering Software Products,
Chapter 1 – Software Engineering, A Practitioner's Approach, 9th Edition

13.3 Learning Unit 3: Software Development Processes and Methodologies

13.3.1 Key Concepts

Software development processes, Software Development Life Cycle, Software Development Models: Traditional Models, Software Development Models: Agile, Agile Methods, Extreme programming, Scrum.

13.3.2 Learning Outcomes and Assessment Criteria

- Apply core software engineering practices at the conceptual level for a given problem.

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- Compare and contrast software development methodologies with respect to environmental, organizational, and product constraints.
- Propose a methodology best suited for a given situation.
- Compare the traditional waterfall model to the agile model, highlighting their distinctive features and assessing the advantages and disadvantages of each.
- Discuss agile methods, identify associated issues, and formulate solutions to mitigate these issues effectively.
- Explain the software lifecycle and its various phases, emphasizing the interdependencies between phases.
- Describe the deliverables produced at each phase and apply this understanding to a practical context.
- Apply an appropriate agile model to solve a given problem, demonstrating the practical application of agile principles in a specific context.

Reference:

- https://www.coursera.org/learn/software-processes#syllabus
- Chapter 1 Introduction to Software Engineering | Coursera
- Chapter 2 – 3 Software Engineering: A Practitioner's Approach, 9th Edition

13.4 Learning Unit 4: Requirements, Features, Scenarios and Stories

13.4.1 Key Concepts

- Software Requirements
- Personas
- Scenarios
- User stories
- Feature identification

13.4.2 Learning Outcomes and Assessment Criteria

Demonstrate a detailed understanding of the modern approach to capturing and recording software requirements in a project.

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- Create personas that vividly represent diverse user types, considering demographics, behaviours, and goals.
- Develop a product vision from a real-world problem.
- Develop comprehensive scenarios covering a range of user tasks and potential situations.
- Integrate personas, scenarios, user stories, and feature identification cohesively to guide product development.
- Apply software features, scenarios, and stories to selected project.
- Apply software features, scenarios, and stories to selected project.

Reference:

- Chapter 3: Engineering Software Products by Ian Sommerville
- Chapter 4 - Software Engineering: A Practitioner's Approach, 9th Edition

13.5 Learning Unit 5: Software Architecture

13.5.1 Key Concepts

- Why is architecture important?
- Architectural design
- System decomposition
- Distribution architecture
- Technology issues

13.5.2 Learning Outcomes and Assessment Criteria

Explore the role of software architecture in software product development. Describe what constitutes software architecture.

- Explain why the architectural design of software is essential.
- Analyse how software architecture affects non-functional requirements.
- Describe and evaluate how software systems are decomposed.

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- Explore different ways of how architectures are distributed.
- Explore technology issues when deciding on software architecture.
- Recommend software architectures given software requirements.

Reference:

- Chapter 4 – Engineering Software Products by Ian Sommerville
- Chapters 9 and 10 -Software Engineering: A Practitioner's Approach, 9th Edition

13.6 Learning Unit 6: Cloud-based Software

13.6.1 Key Concepts

- Virtualisation
- Containers
- Everything as a service
- Software as a service
- Multitenant and multi-instance systems
- Cloud software architecture

13.6.2 Learning Outcomes and Assessment Criteria

- Cloud Computing Fundamentals: Demonstrate a comprehensive understanding of the concepts and principles underlying cloud computing, including its definition, characteristics, benefits, and challenges.
- Virtualization and Containers: Explain the concepts of virtualization and containerization, differentiating between them and discussing their role in cloud-based software delivery. Analyze the advantages and disadvantages of each approach.
- "Everything as a Service" (XaaS): Describe and differentiate between various XaaS offerings, including Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). Provide real-world examples of each and discuss their applicability in different scenarios.

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- Multi-tenancy and Multi-instance Architectures: Elaborate on the implementation and implications of multi-tenancy and multi-instance architectures in cloud environments. Compare and contrast these approaches, considering factors such as security, scalability, and cost-effectiveness.
- Cloud-Based Software Delivery: Apply the knowledge of cloud computing concepts to explain how software products are developed, deployed, and delivered via the cloud. Discuss the advantages and disadvantages of this approach compared to traditional software delivery methods.
- Assessment Criteria
    - Demonstrating comprehension of key concepts, comparing and contrasting different approaches, and applying knowledge to specific scenarios. Examples include explaining the trade-offs between IaaS, PaaS, and SaaS, or analyzing the security implications of multi-tenancy.
    - Hands-on activities involving the use of cloud platforms or tools to deploy and manage simple applications. This could include setting up a virtual machine, deploying a containerized application, or using a cloud-based database service.
    - Presenting research or analysis on specific cloud-related topics, demonstrating clear communication and in-depth understanding of the subject matter. This could involve comparing different cloud providers or exploring emerging trends in cloud computing.
    - Assessing knowledge of key terms, concepts, and principles related to cloud-based software. This could include multiple-choice questions, short-answer questions, or essay-style questions.
- Reference
    - Chapter 5 – Engineering Software Products by Ian Sommerville.
    - Check WiSeUp for more information.

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14 Scheme of Work

WEEK(S)	Learning & Teaching Strategies (Methods)	LECTURE/TUTORIAL ACTIVITY	STUDENT ACTIVITY
Week 1	Lecture, discussion, videos	Unit 1 – Software Engineering,	Read Unit one notes. Study chapter 1 from Roger S Pressman Register at Coursera. Participate in Discussions posted on Moodle, Register at IBM Watch the Video on LLM and End of Programming. Assignment 1 Released.

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Week 2	Lecture, discussion, videos	Unit 1 – Software Engineering continues. (SE Stakeholders, Legal, ethics and professional issues	Complete Unit 1 study notes on Code of Ethics and Professional Issues in Software. Identify dilemmas that SE can find themselves in, Study Assignment 1 Quiz 1 Released.
Week 3	Lecture, discussion, videos, class activity	Unit 2 – Software Products	Study Unit 2 notes from the textbook. Identify types of software Discuss the findings from Video on LLM and End of Programming

Week 4	Lecture, discussion, videos, lab	Unit 2 – Software Products (continues)	Study notes. Quiz 2 released

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Week 5	Lecture, discussion, videos, lab	Unit 3 Software Development Processes & Methodologies • Requirements and Architecture. • processes that are used to implement the software, verify, and validate the software, deploy the software and maintain the software	Complete registration with Coursera. Watch videos on Requirements and architecture. Do short quizzes on Coursera
Week 6	Lecture, discussion, videos, lab	Unit 3 Software Development Processes & Methodologies (Continues) • Traditional Models and Agile methods • XP and Scrum	Apply exception handling to code. Discussions, Watch Video demos, Practice code Quiz 3
Week 7	Lecture, discussion, videos, lab	Unit 4 Requirements: - Features, Scenarios and Stories • Features, Scenarios, Stories	Study notes. Test 1 Participate in Discussions posted on Moodle

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Week 8	Lecture, discussion, videos, lab	Unit 5 - Software Architecture • What is software engineering? • Why is architecture important? • Architectural design • System decomposition • Distribution architecture • Technology issues	Watch demo videos, Design features for your project. Quiz 4
Week 9	Lecture, discussion, videos, lab	Unit 6- Cloud-based Software	Incorporate software architecture into your project. Quiz 5
Week 10	Lecture, discussion, videos, lab	Unit 6- Cloud-based Software	Experiment with Cloud-based software, Virtualization and docker containers Test 2
Week 11 &12		Project Submission	Revision and preparation for examination

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15. Notes on Plagiarism

Plagiarism constitutes a breach of academic integrity, compromises, and undermines the values and processes by which knowledge is created, shared, and evaluated. Such a breach not only casts suspicion on the integrity of the individual involved but also damages the University's reputation. Walter Sisulu University (WSU) takes serious consideration of its role to uphold its academic integrity and to promote trust in the scholarly and administrative work undertaken at the University. The plagiarism policy serves as an overseer in ensuring the University protects the intellectual property of the originators of the sources of information its stakeholders use to teach, research, learn, and manage.

Plagiarism is the act of submitting or presenting work, study and assessment material, assignments, research work or inventions of someone else, irrespective of its source, as one's own creation; in some instances, even where credit or acknowledgement is given to the original source. Examples of conduct that may fall **within the definition include** using the direct words of another without using quotation marks (even where the passage is referenced), Unacknowledged copying of a sentence, or copying more extensive blocks of text. Syndication of a single piece of work by more than one student (except where the assignment task is a legitimate group assignment. Borrowing or using another person's assignment. Submitting an entire essay from another person or from the internet, or infringing copyright.

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